UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

CS 498: Inference Overview

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Materials with thanks to Minjia Zhang, Arvind Krishnamurthy, and many colleagues

Recall: Distributed Model Training

- **Model execution exceeds the capacity of single machines**
  - Data parallelism, pipeline parallelism, tensor parallelism

- **Reduce the resource demand of training**
  - Mixed precision training (e.g., INT8)
  - Gradient accumulation
  - Activation checkpointing
  - Low-rank adaptation

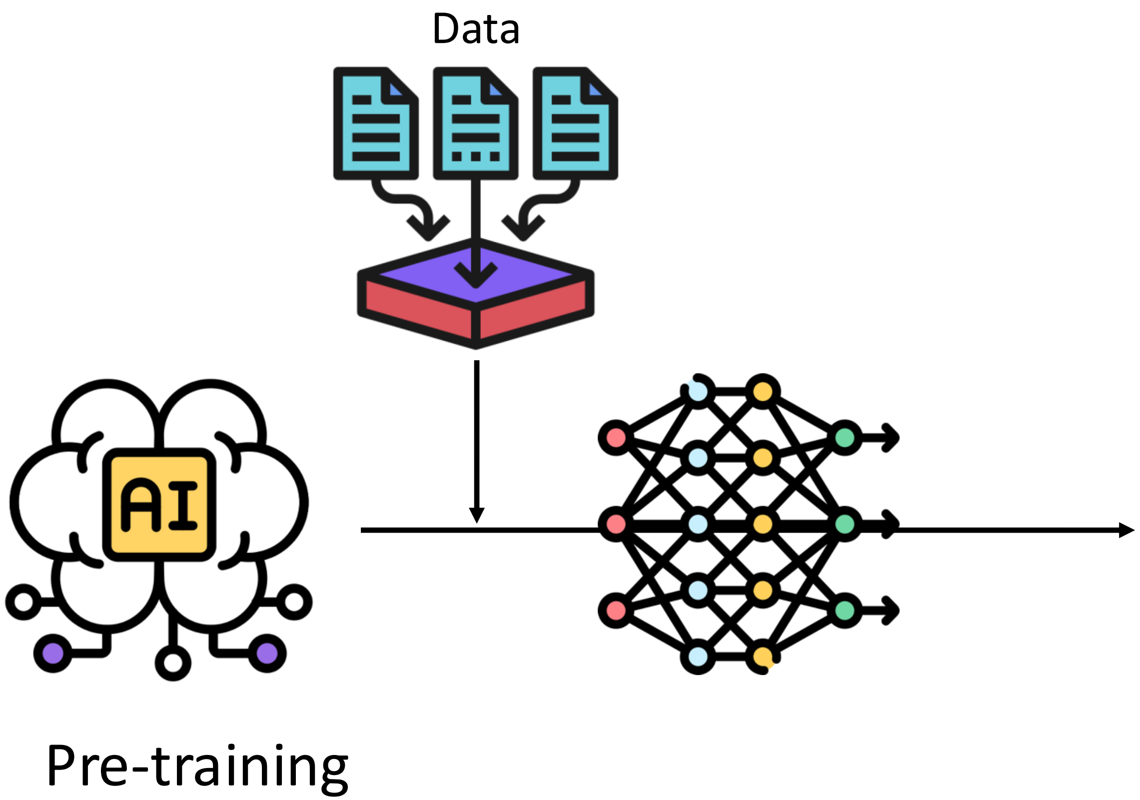
Today

LLM Inference Overview

- Background
- Challenges
- System optimizations
- Algorithm optimizations

Panel Discussion (Speculative Decoding)

What is Model Inference/Serving?



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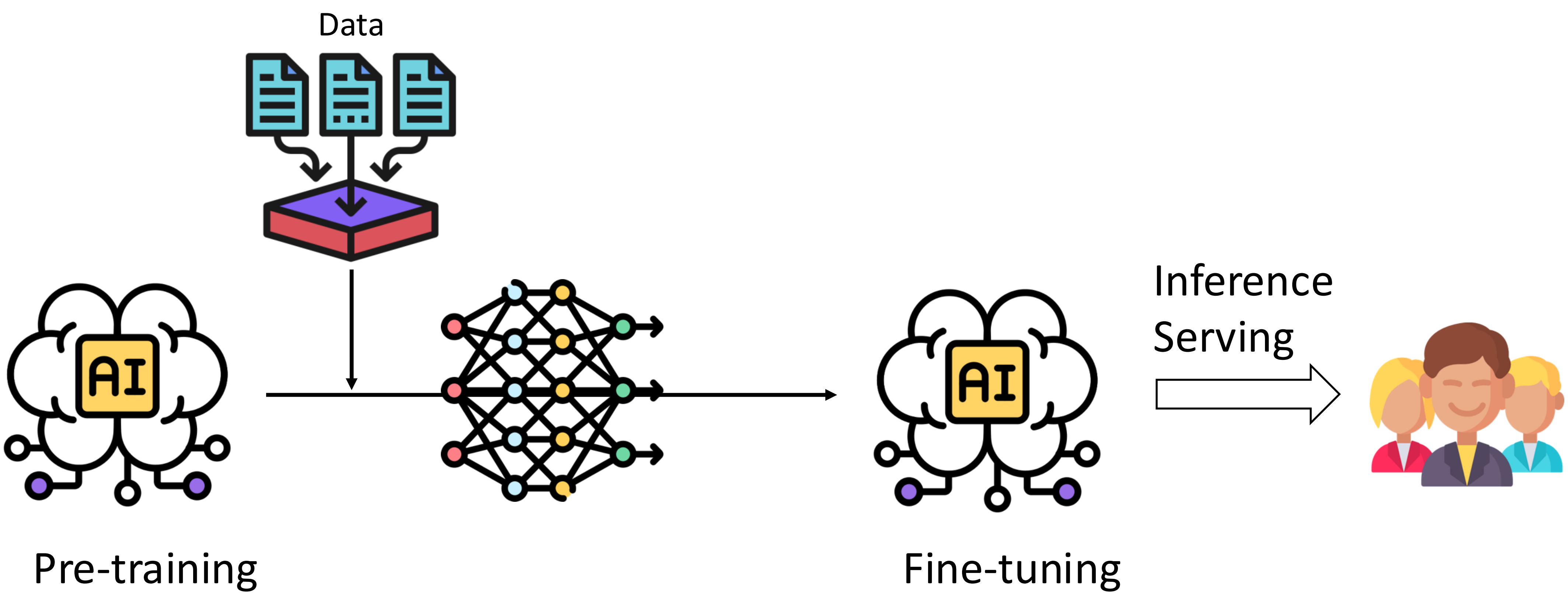
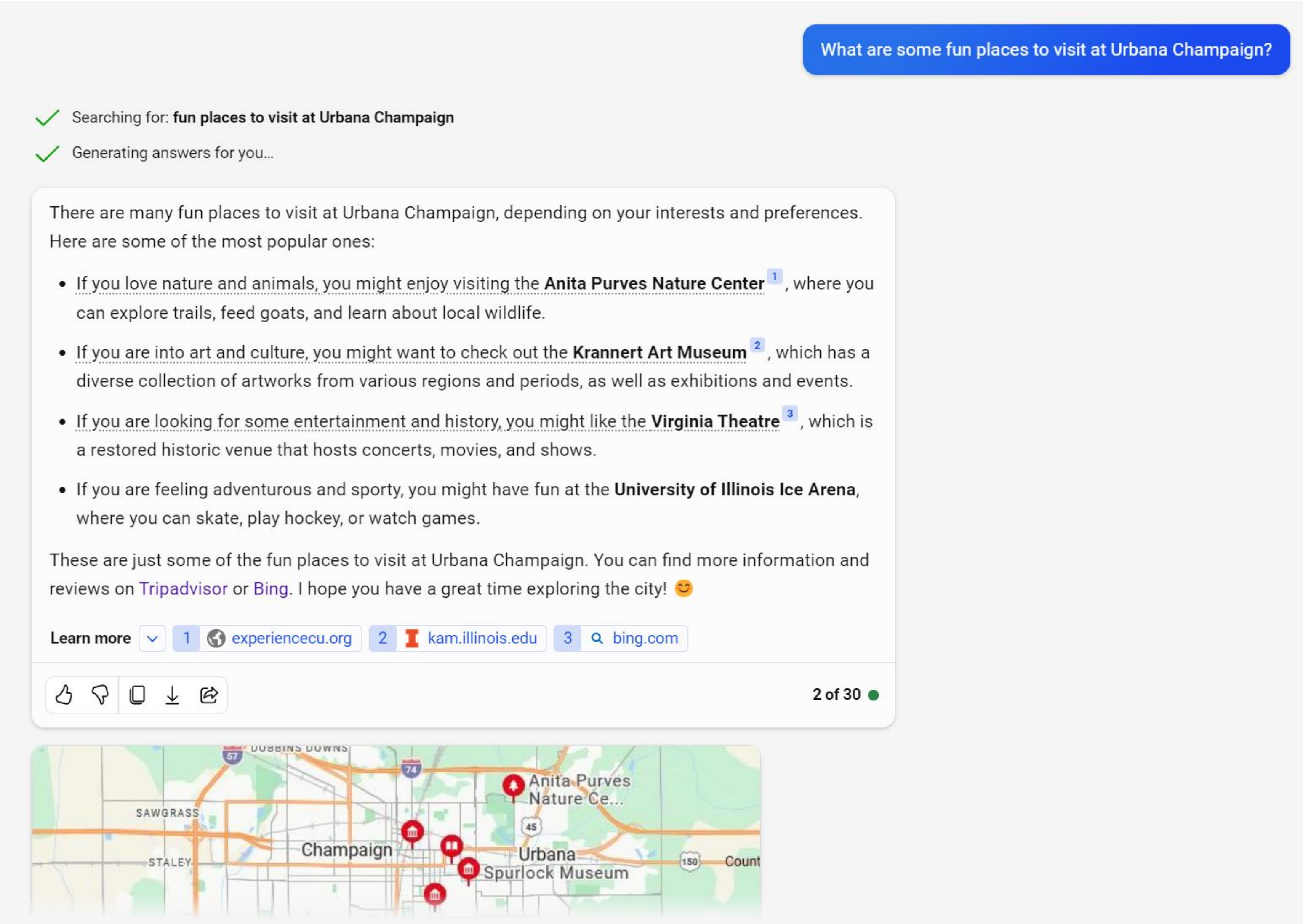


Inference Scenario 1: ChatGPT/SearchGPT/Copilot



Microsoft Copilot: Your AI companion

Inference Scenario 2: Online Image/Video Generation

DALL·E 3 · An expressive oil painting of a basketball player dunking, depicted as an explosion of a nebula.

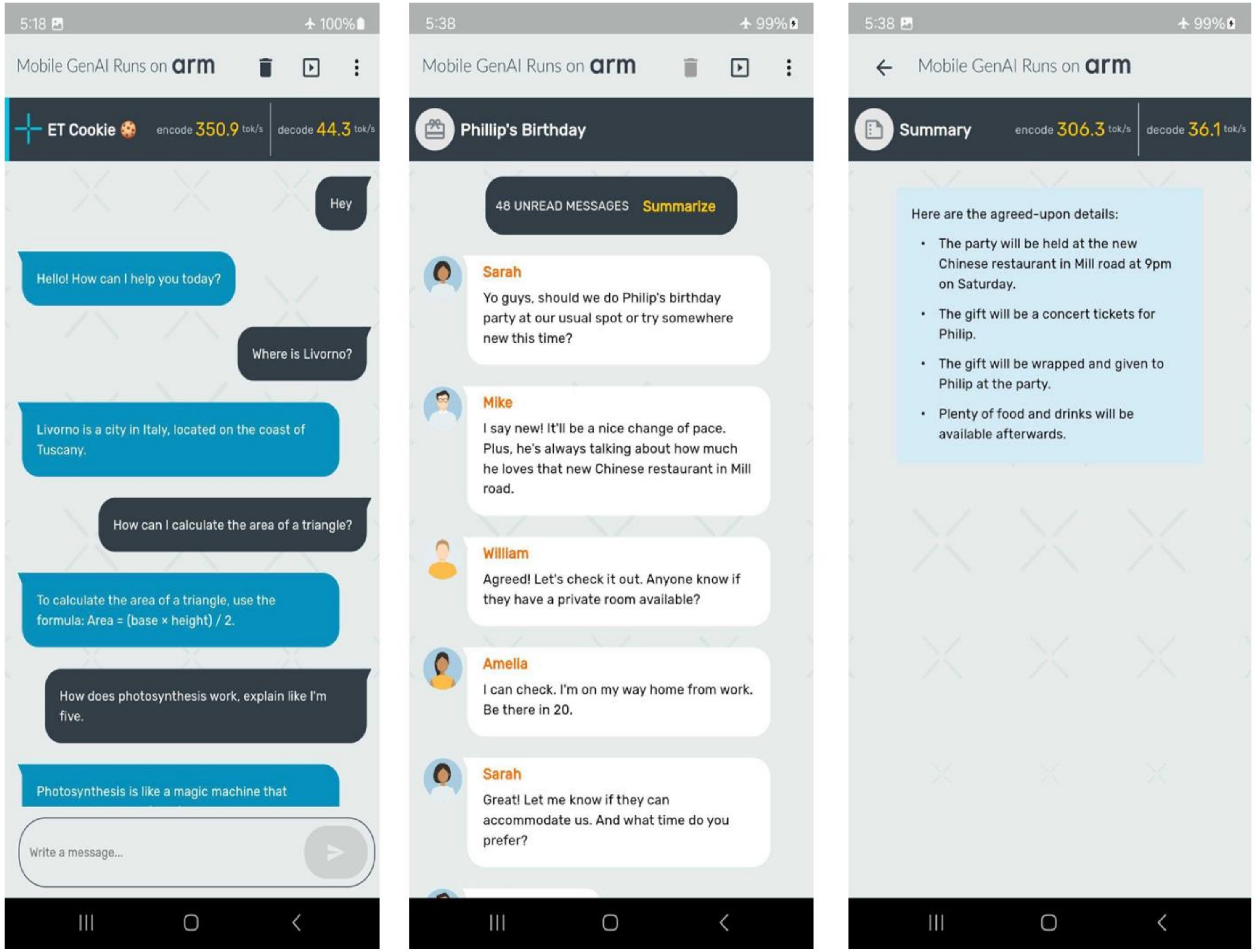
DALL·E 3 | OpenAI

Prompt: A stylish woman walks down a Tokyo street filled with warm glowing neon and animated city signage. She wears a black leather jacket, a long red dress, and black boots, and carries a black purse. She wears sunglasses and red lipstick. She walks confidently and casually. The street is damp and reflective, creating a mirror effect of the colorful lights. Many pedestrians walk about.

Sora2: Creating video from text

Inference Scenario 3: Surveillance System

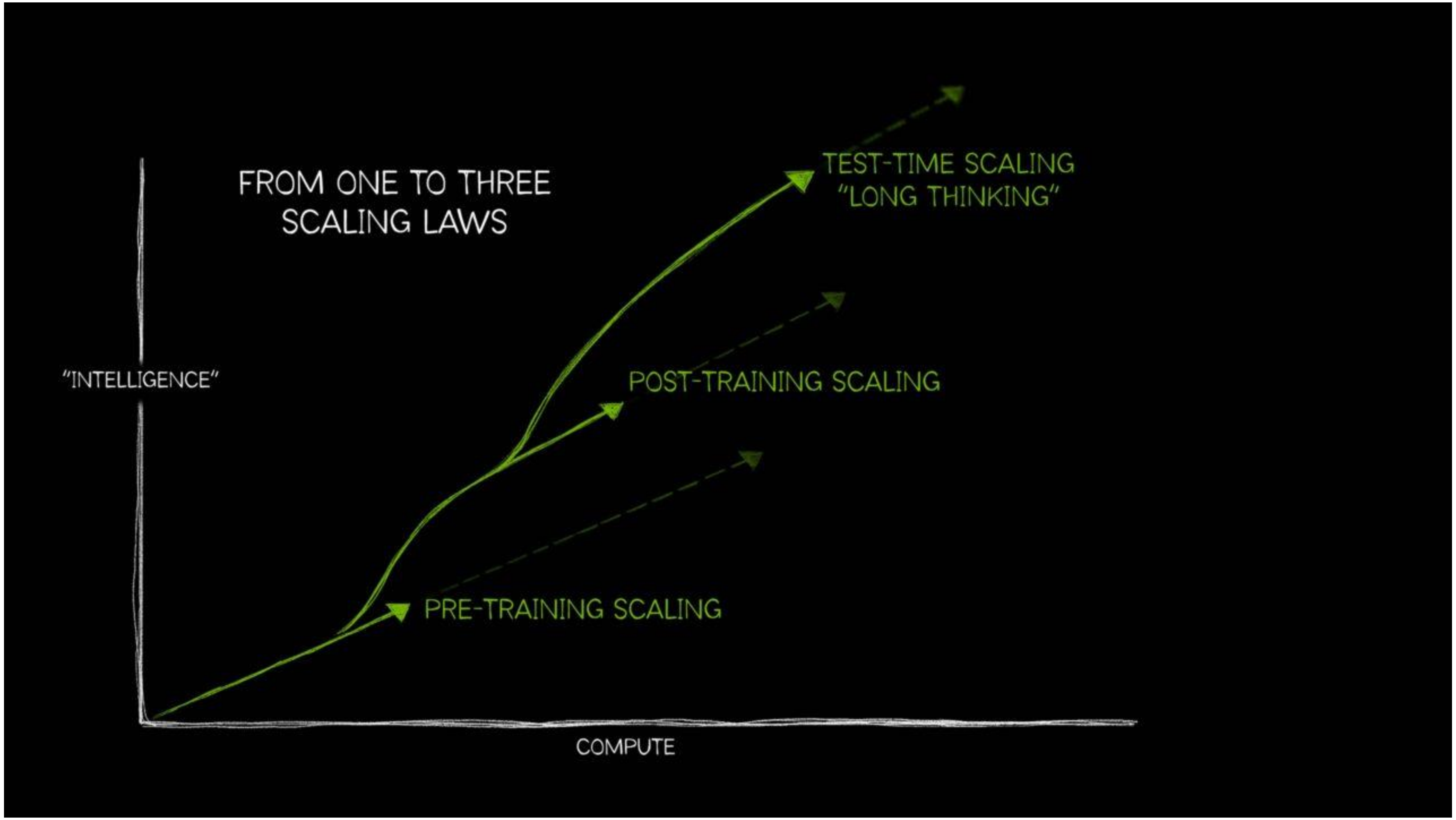
Inference Scenario 4: Mobile Apps



[Unleashing the Power of AI on Mobile: LLM Inference for Llama 3.2 Quantized Models with ExecuTorch and KleidiAI | PyTorch]

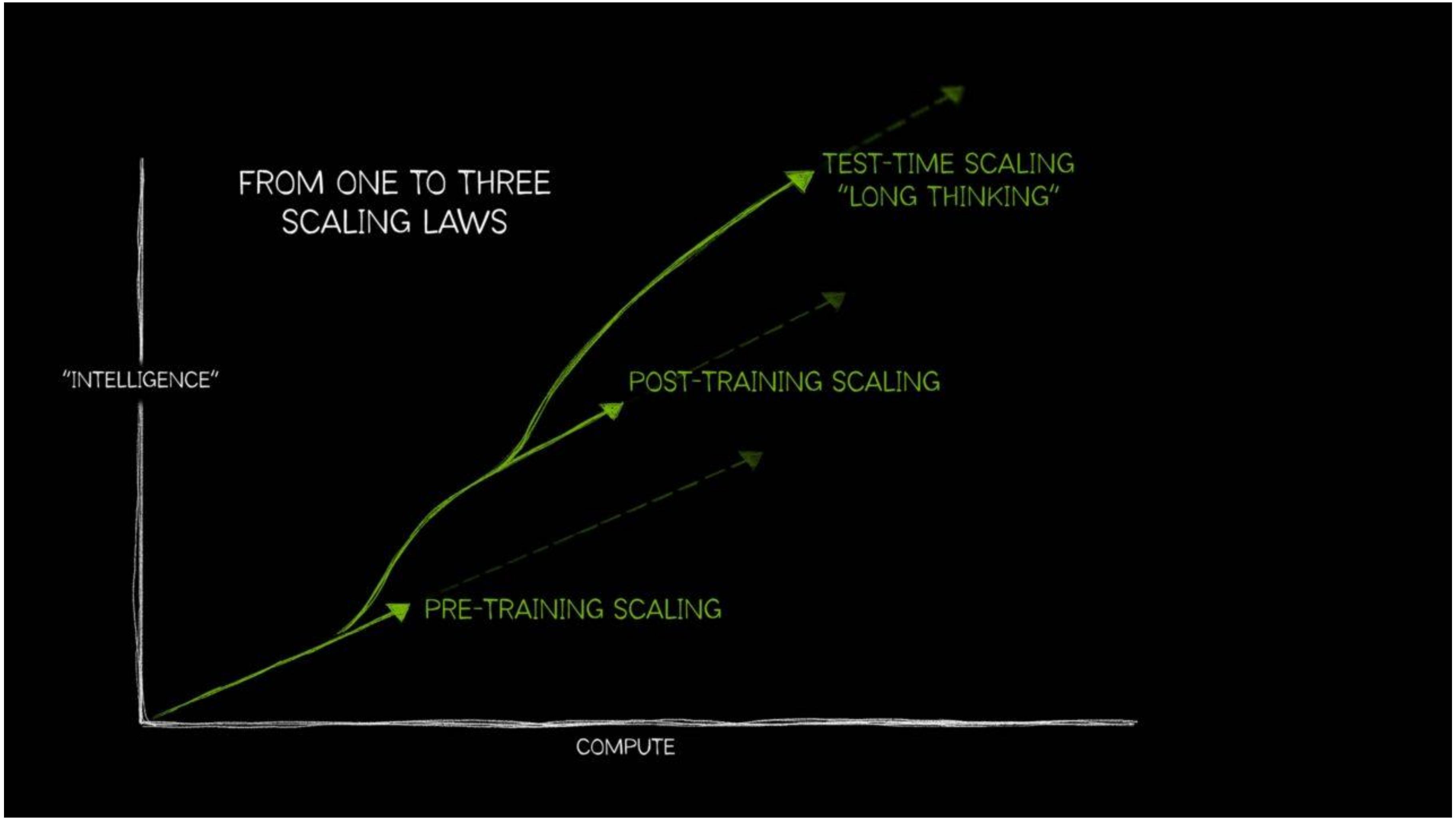
Inference Scenario 5: Augmented and Virtual Reality

Inference Time Scaling



NVIDIA Jensen Huang at GTC 2025

Inference Time Scaling



1 Pretraining scaling = bigger model and data, better performance

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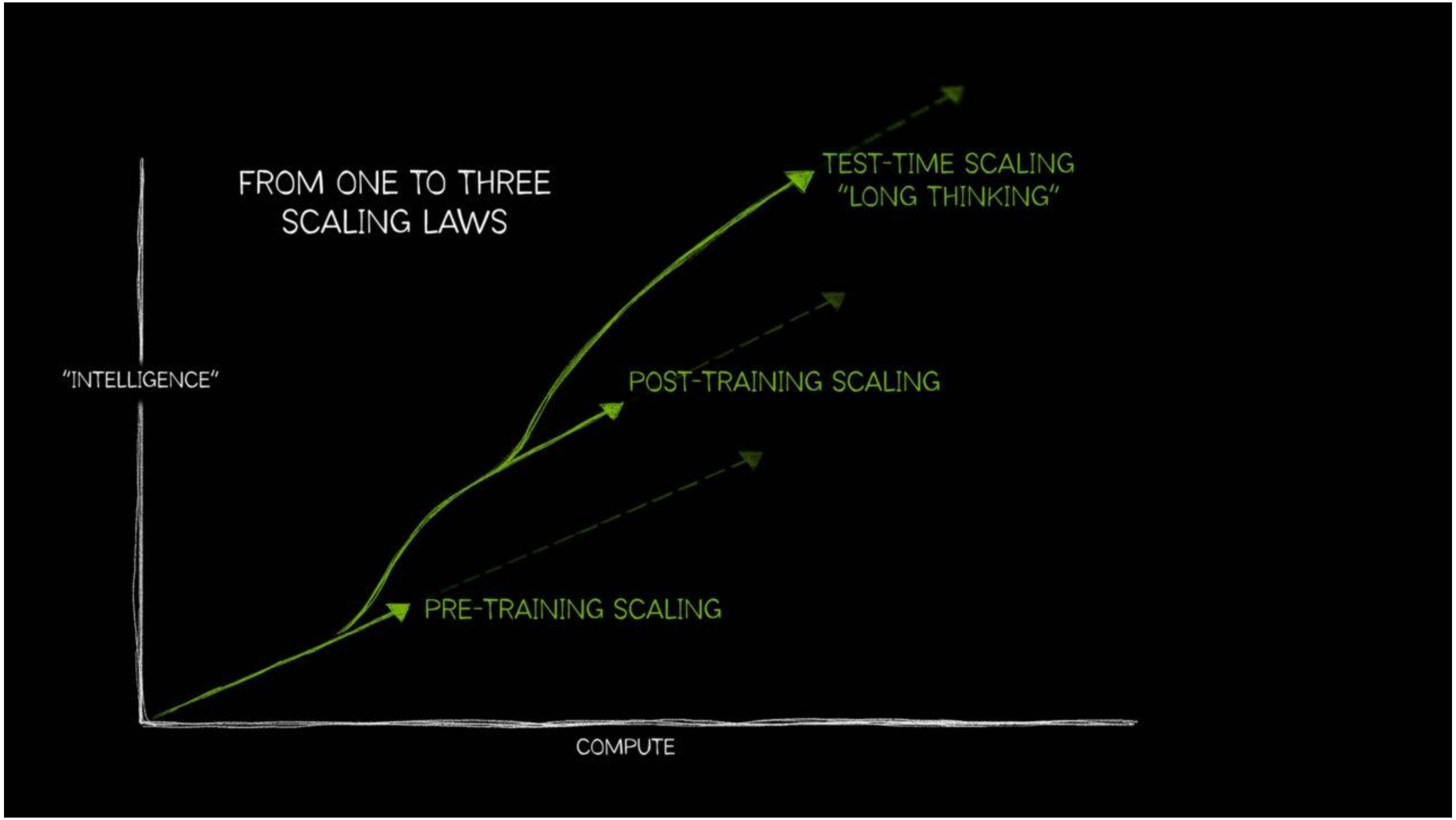
Inference Time Scaling

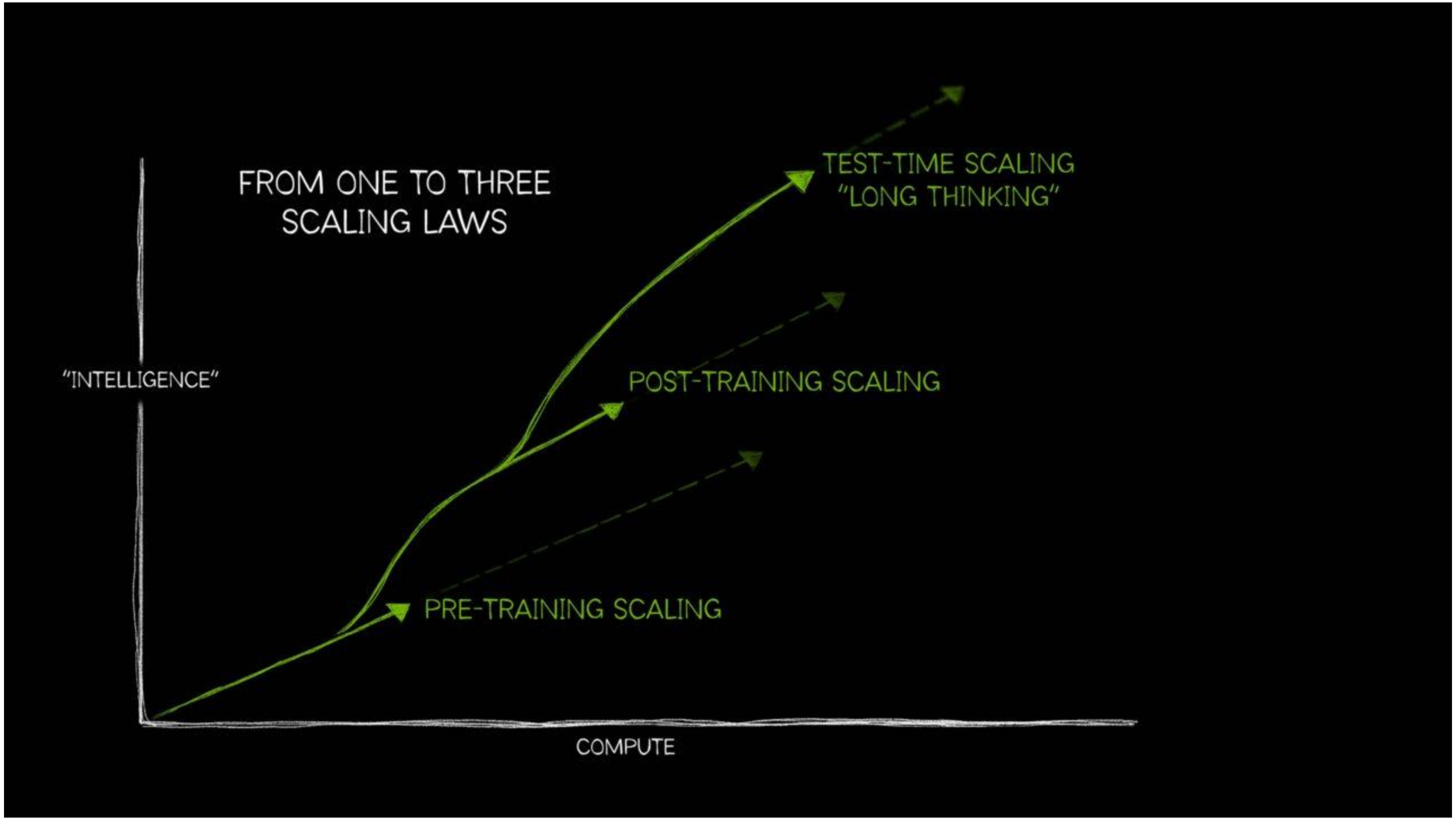


1 Pretraining scaling = bigger model and data, better performance

2 Post-training scaling = fine-tuning for precision

NVIDIA Jensen Huang at CES 2025

Inference Time Scaling Requires More Compute



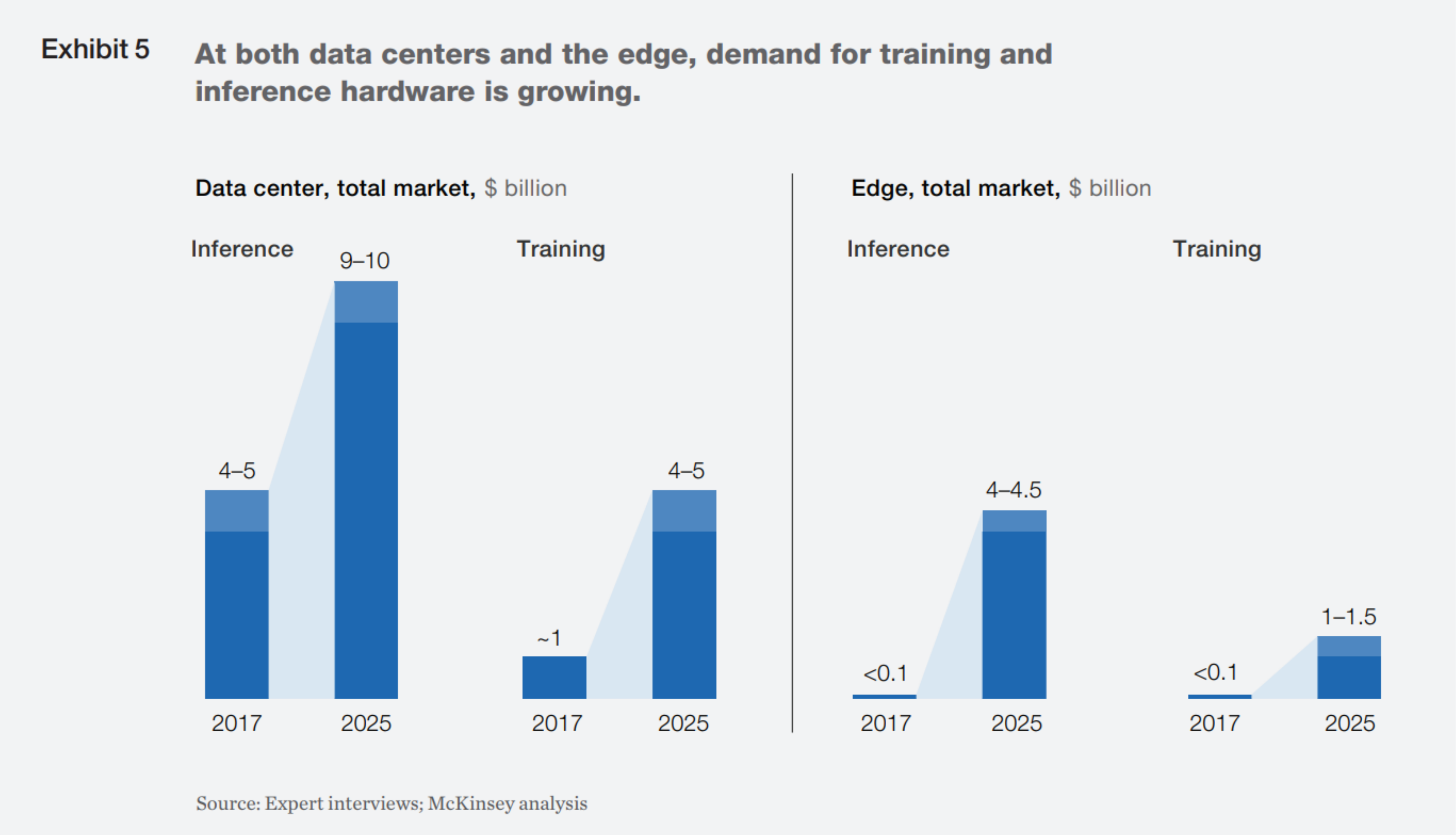
1 Pretraining scaling = bigger model and data, better performance

2 Post-training scaling = fine-tuning for precision

3 Test-time scaling (long thinking) = multi-pass reasoning for complex problems/try it many times

NVIDIA Jensen Huang at CES 2025

Training -> Inference



Training vs. Inference

	Training	vs	**Inference**
Runtime	Weeks or months		Milliseconds or seconds

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Challenges	Speed, Scale, Data		SLA (Service-level agreement), TCO (total cost of ownership)

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SLA (LLM: token rates)

TCO

- Parameter volume
- LLM: Context length

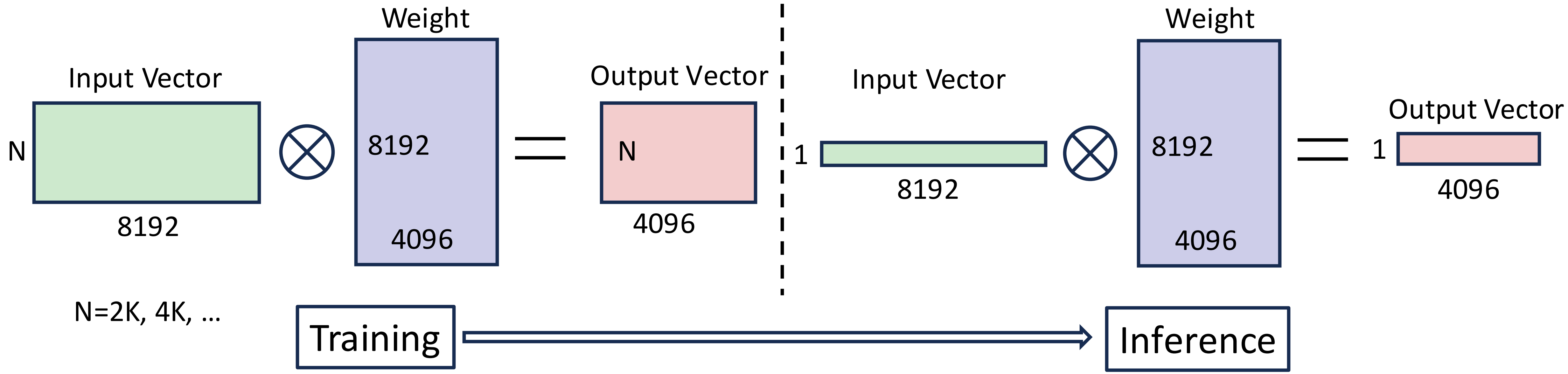
Inference Challenge 1: Long Latency Violates SLA

- Long serving latency blocks deployment
- Support advance models while meeting latency SLA and saving cost

DL Scenarios	Original Latency	Latency Target
Deep Query Document Similarity	10~12ms for [query, 1 doc] x 33 docs	< 6ms
Travel Planning	10ms for [query, 1 passage] x 150 passages	< 5ms
Ads seq2seq model for query rewriting	~51ms	< 5ms

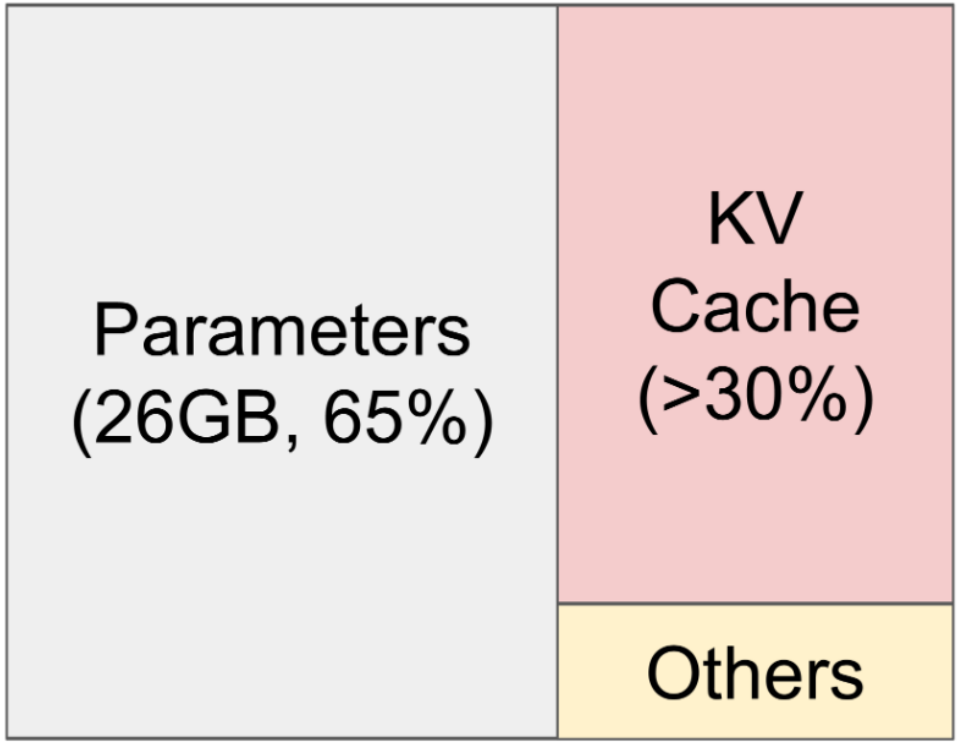
Inference Challenge 2: Small Batch Limits Parallelism

- Small batch size $\Longrightarrow$ Low latency yet low data reuse (arithmetic intensity)
- Autoregressive generation $\Longrightarrow$ Generate tokens one-by-one



Inference Challenge 3: Large Memory Increases Cost

- Model parameters
  - # Layers
  - # Hidden dim
- KV cache (in attention layer)
  - Batch size
  - Sequence length
  - # Layers
  - # Hidden
- Activation and others



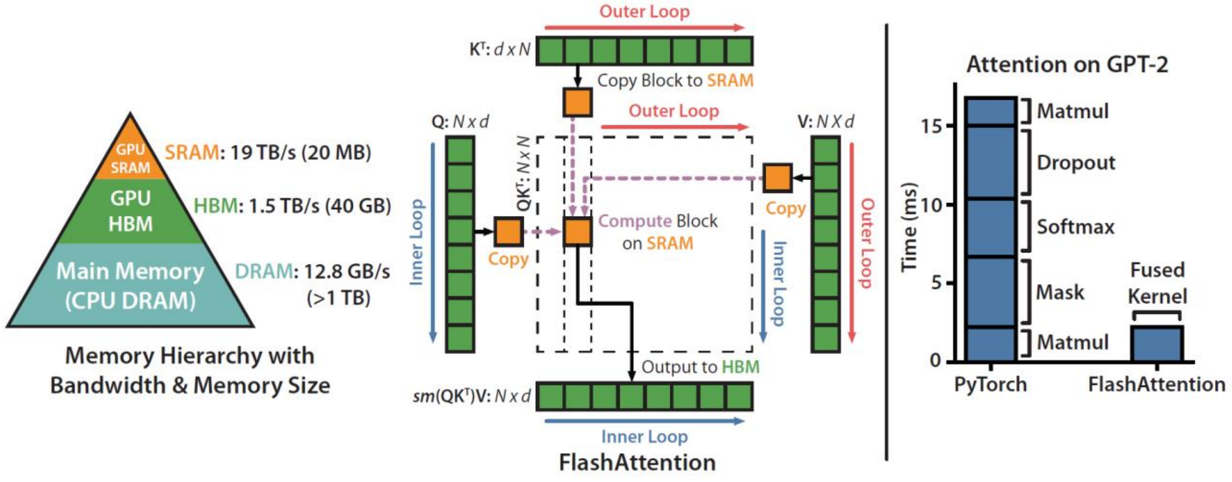
LLaMA-13B on A100 40 GB

Efficient Memory Management for Large Language Model Serving with PagedAttention, by Kwon et al., 2023

Recent Advances to Tackle These Inefficiencies

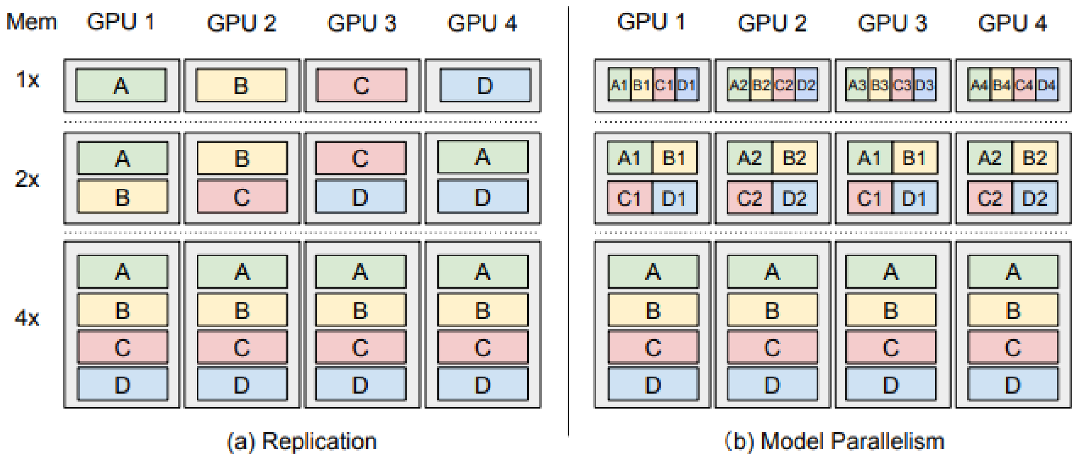
- From a down-to-top view
  - Execution layer (e.g., how to execute efficiently)
  - Planning layer (e.g., how to schedule resources)
  - Algorithm layer (e.g., how to change model)

Customized Kernels & Model Parallelism

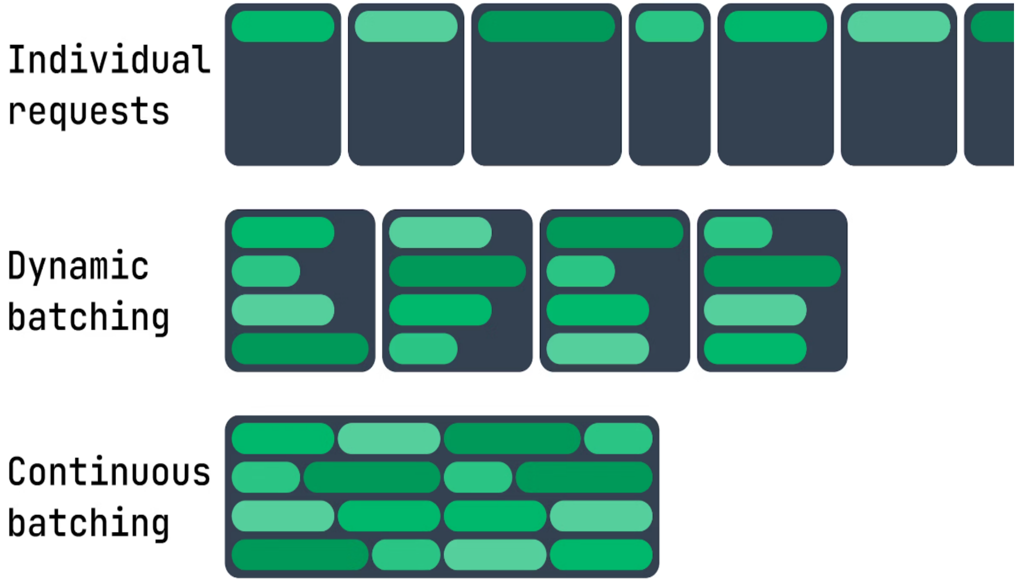


FlashAttention: Fast and Memory-Efficient Exact Attention with IO-Awareness, 2022

AlpaServe: Statistical Multiplexing with Model Parallelism for Deep Learning Serving, OSDI 2023

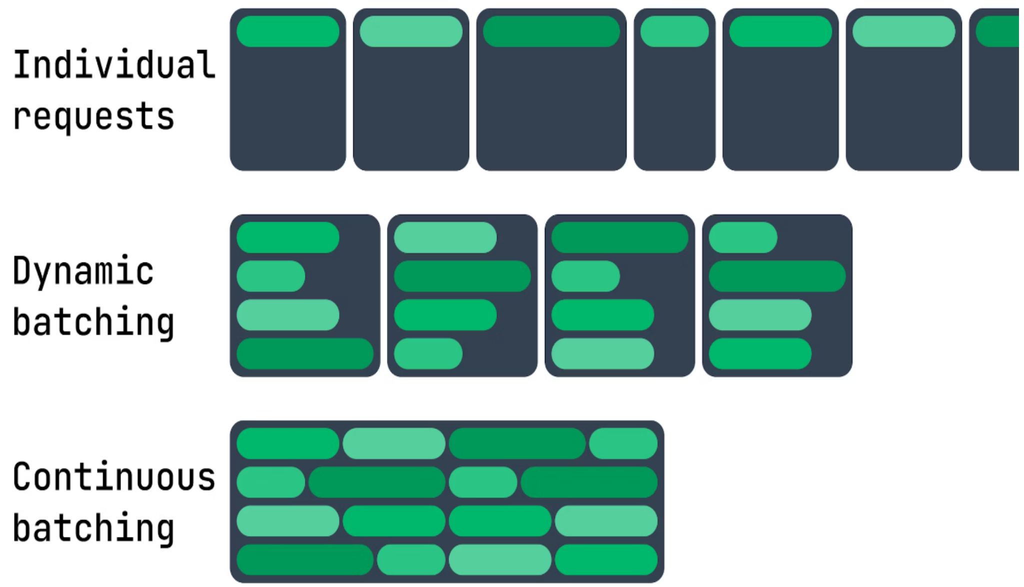


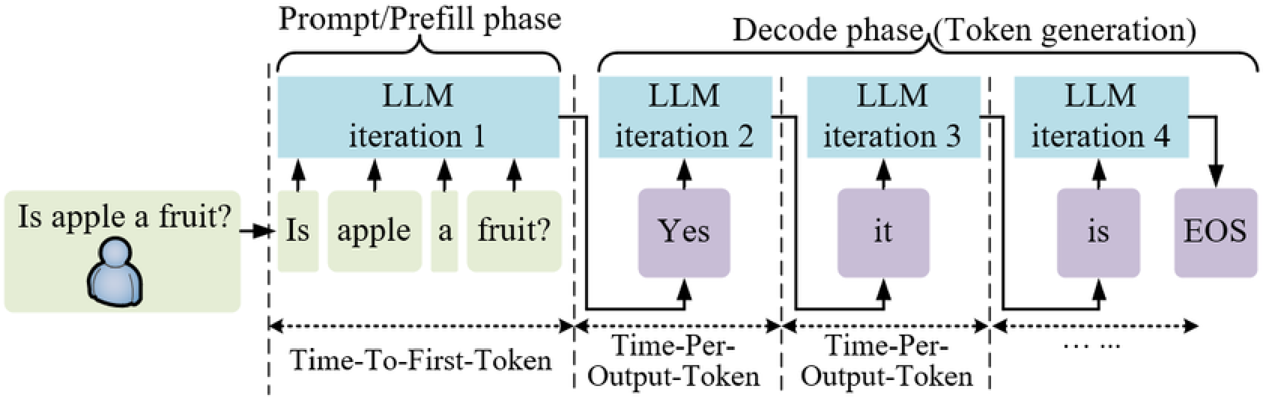
Scheduling Strategies for LLM Inference



Orca: A Distributed Serving System for Transformer-Based Generative Models, OSDI 2022

Scheduling Strategies for LLM Inference



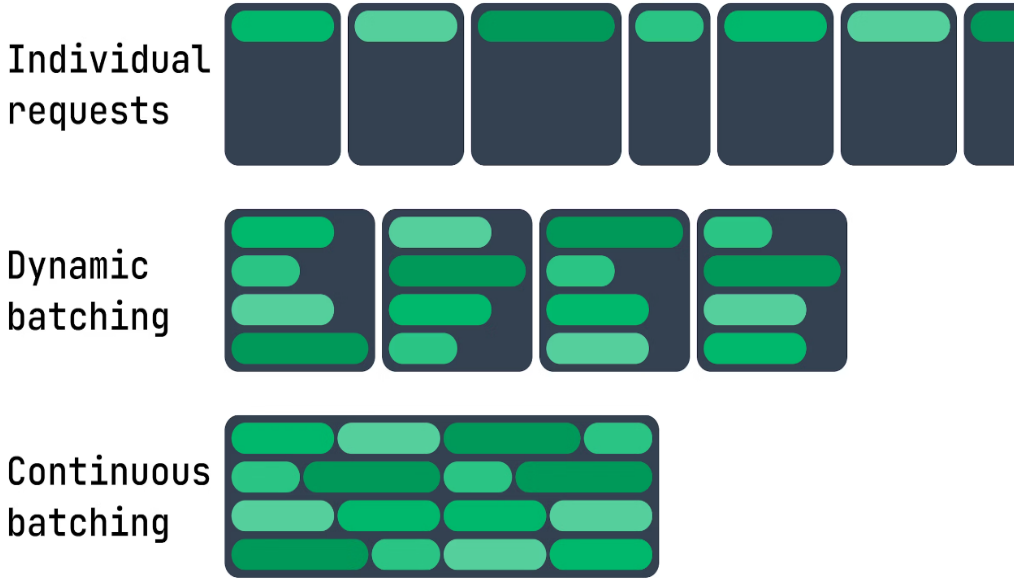


Orca: A Distributed Serving System for Transformer-Based Generative Models, OSDI 2022

LLM generation is autoregressive:

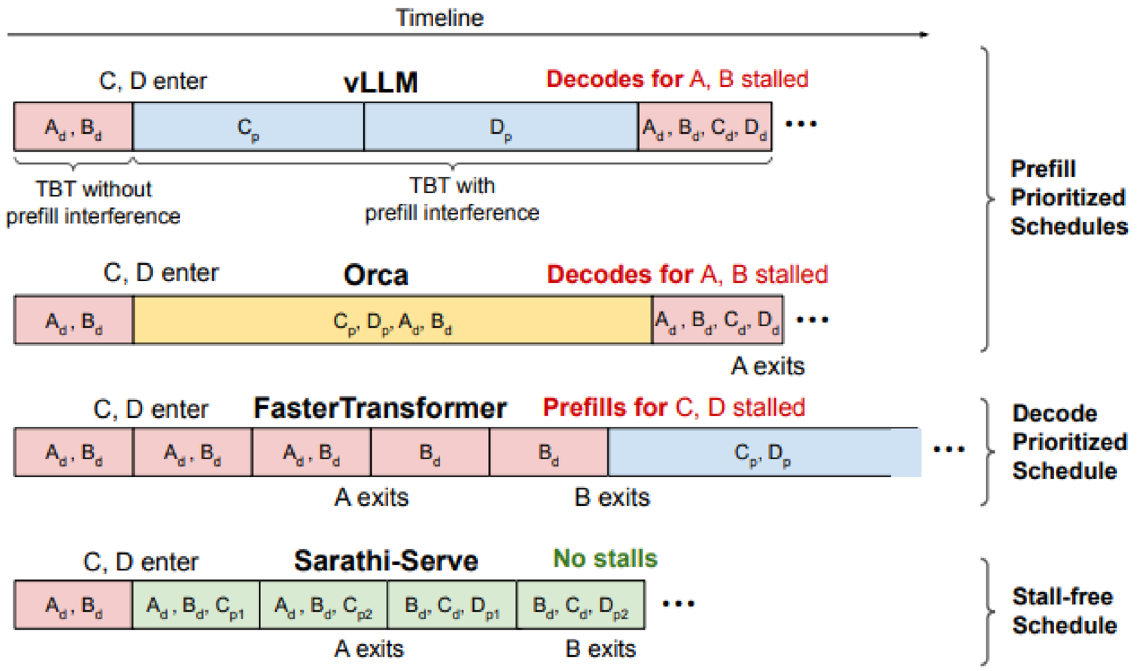
- Prefill stage: digest all input
- Decode stage: digest tokens by tokens

Scheduling Strategies for LLM Inference

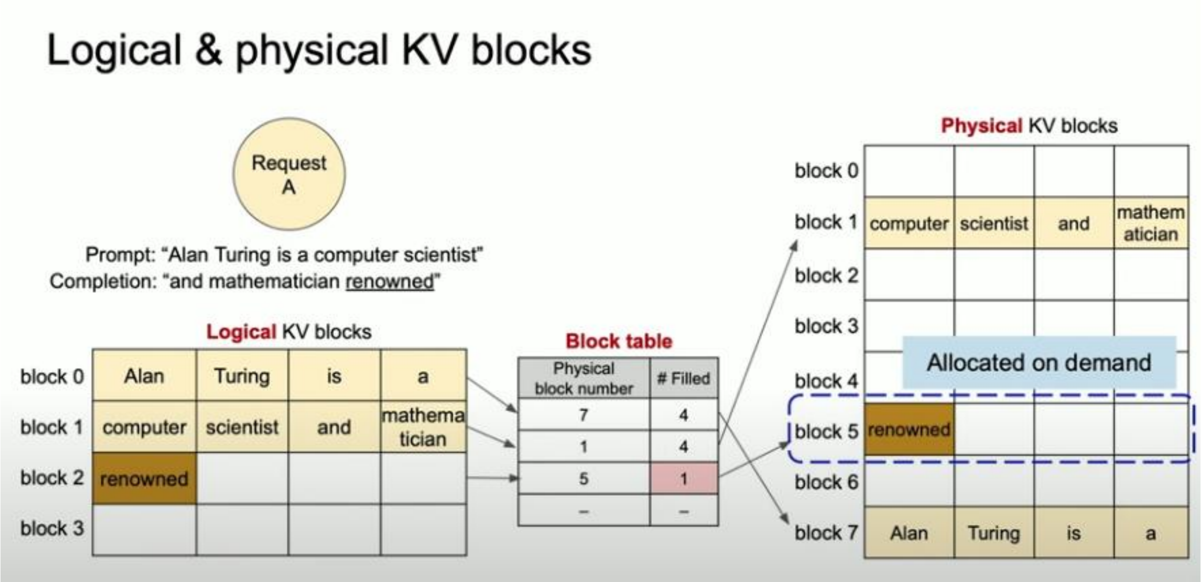


Orca: A Distributed Serving System for Transformer-Based Generative Models, OSDI 2022

Taming Throughput-Latency Tradeoff in LLM Inference with Sarathi-Serve, OSDI 2024

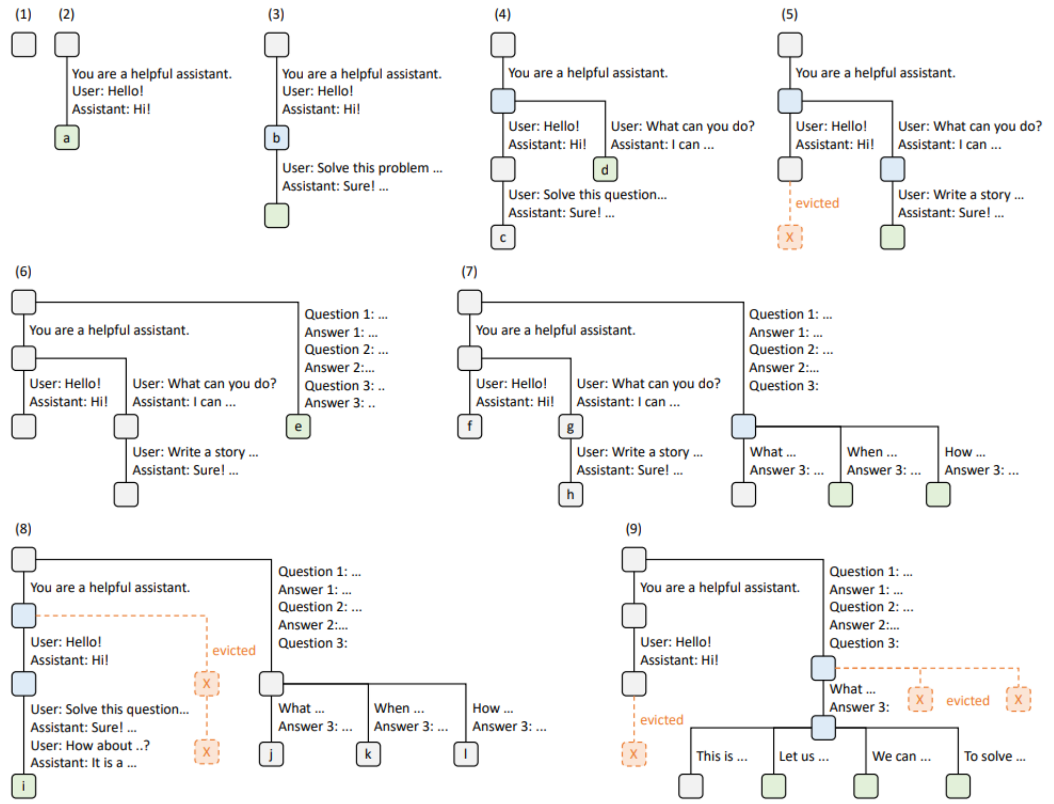


KV Cache Management



Efficient Memory Management for Large Language Model Serving with PagedAttention, 2023

SGLang: Efficient Execution of Structured Language Model Programs, 2024



Recent Advances to Tackle These Inefficiencies

- From a down-to-top view
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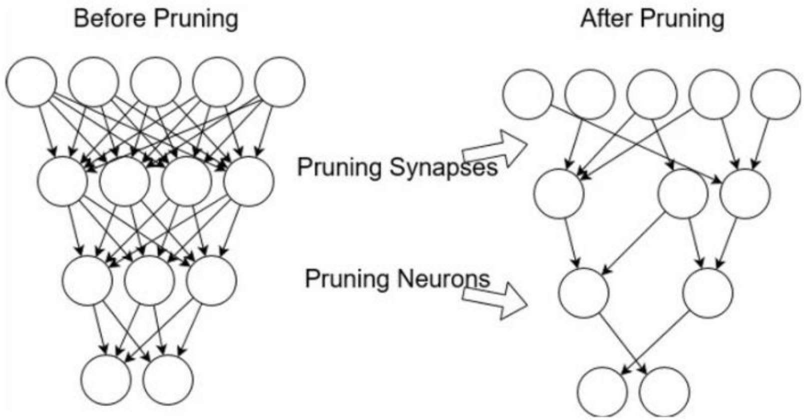
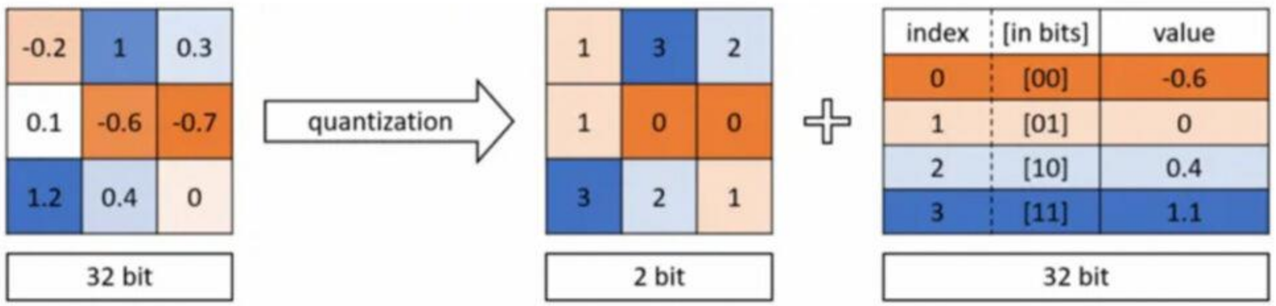
Compression Strategies

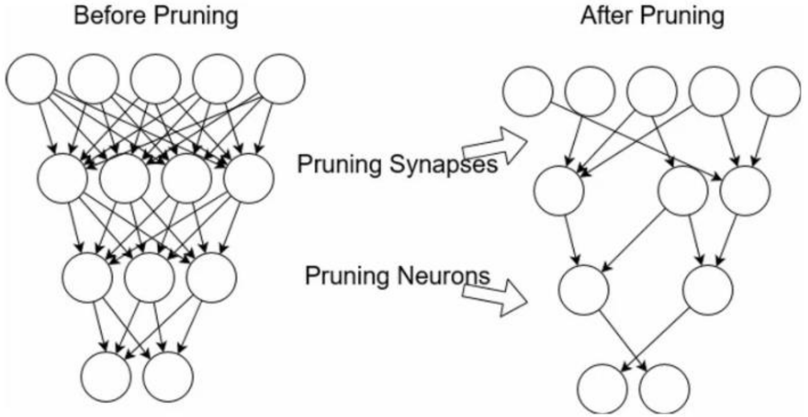


Pruning/Sparsification



Quantization

Compression Strategies



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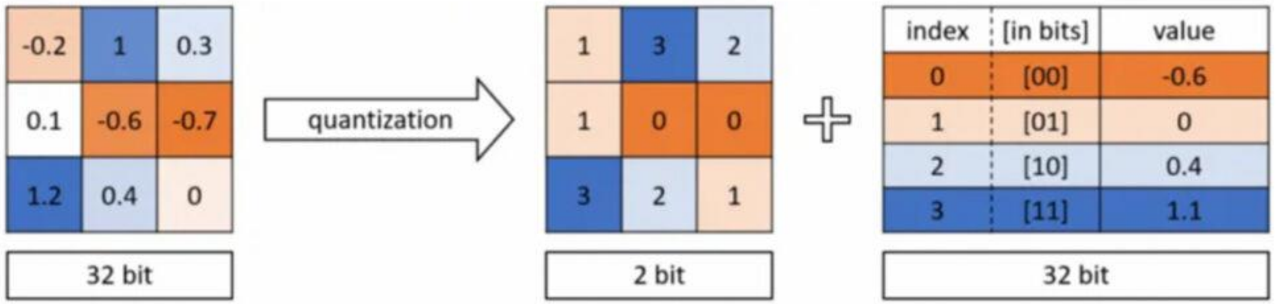

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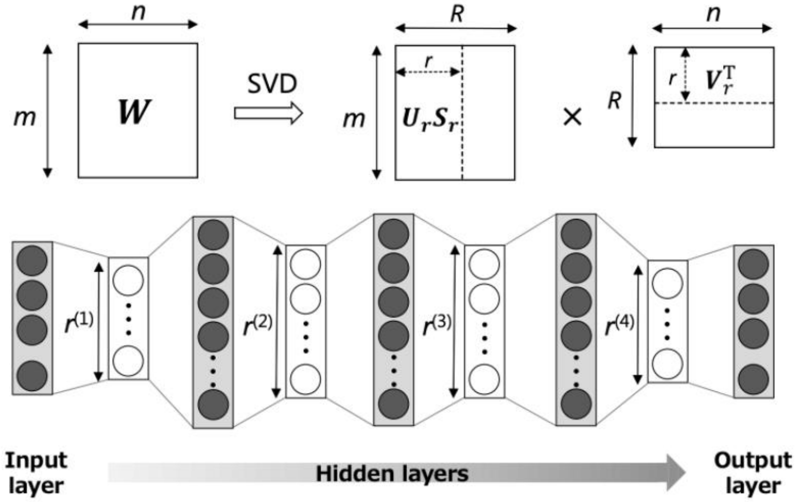
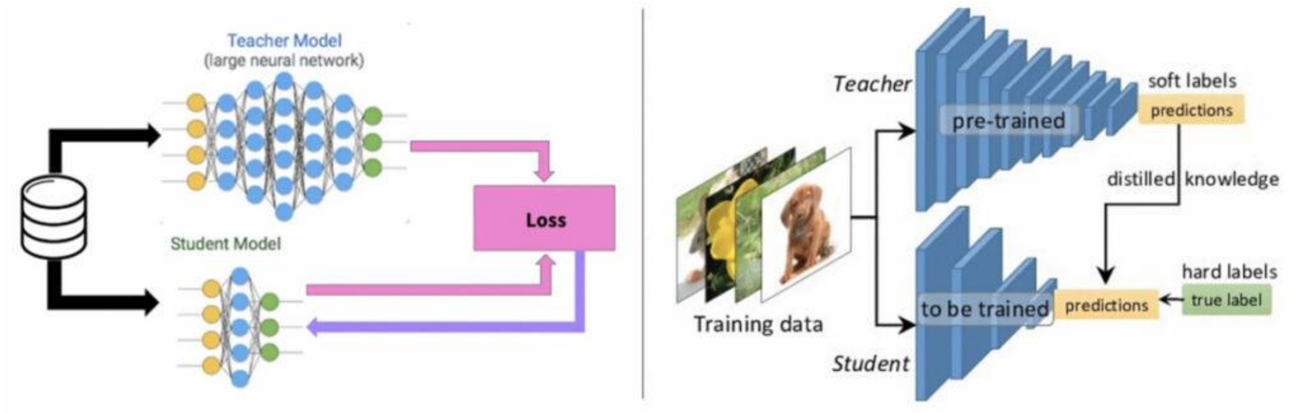


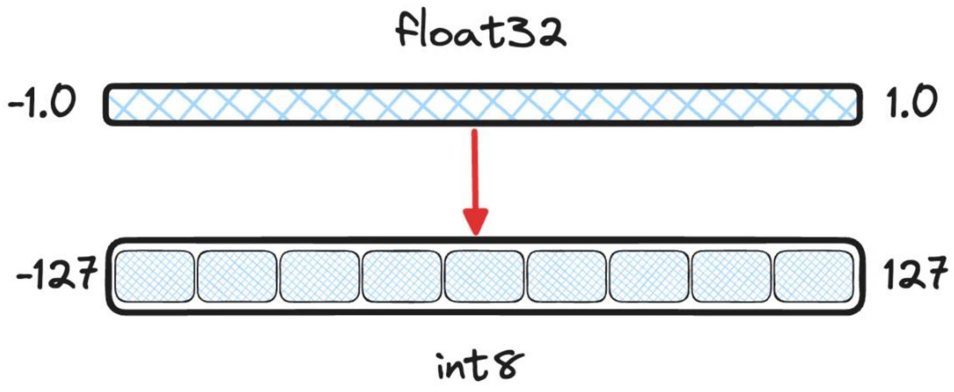
Low-rank decomposition



Distillation

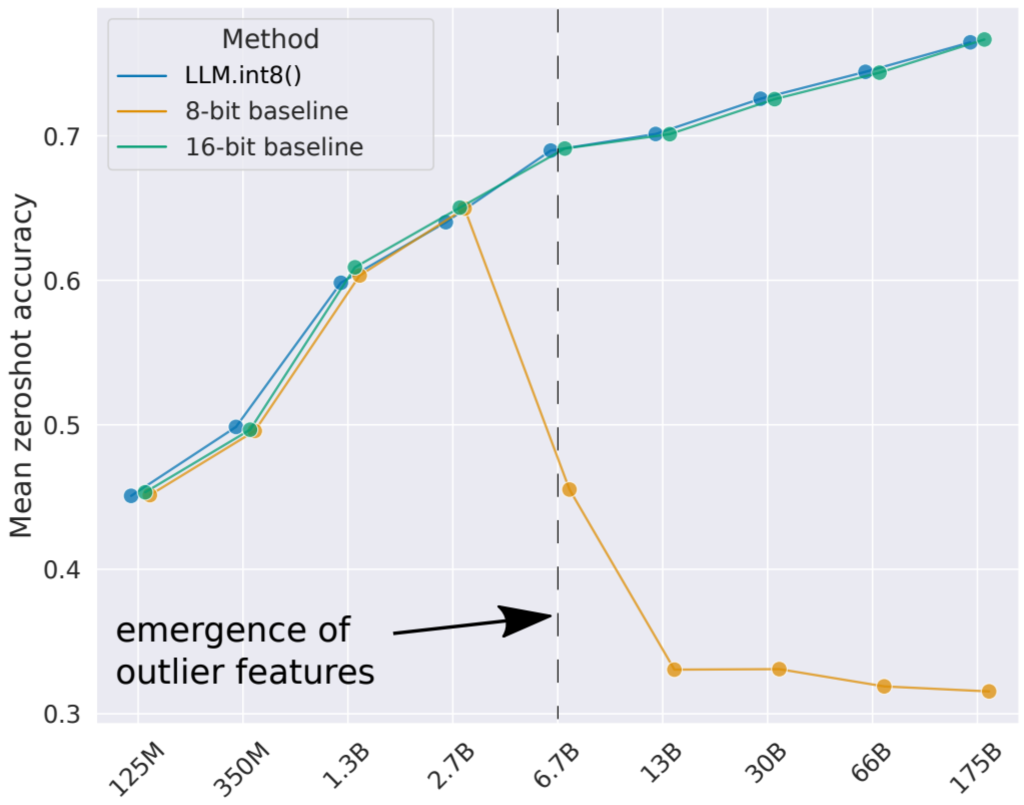
Quantization: Quick Recap

- Reduce the bits per weight, saving memory consumption
- Accelerate inference speed on supporting hardware



Challenges to Quantize LLMs

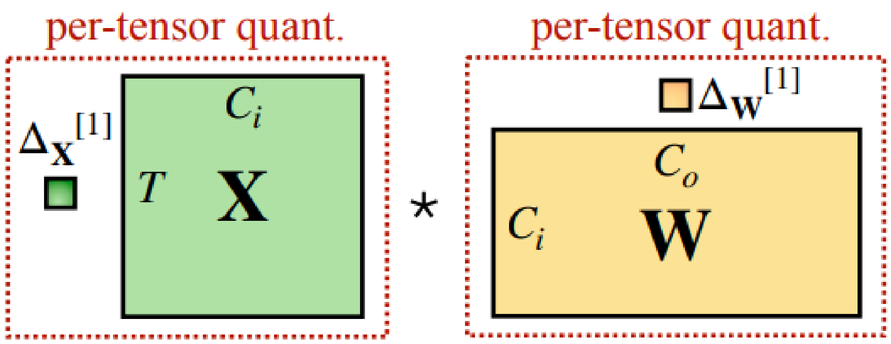
- Standard quantization strategy leads to catastrophic accuracy drop



LLM.int8(): 8-bit Matrix Multiplication for Transformers at Scale, 2023

Fine-grained Quantization

- Per-tensor quantization
  - Low accuracy
  - Fast to quantize/dequantize

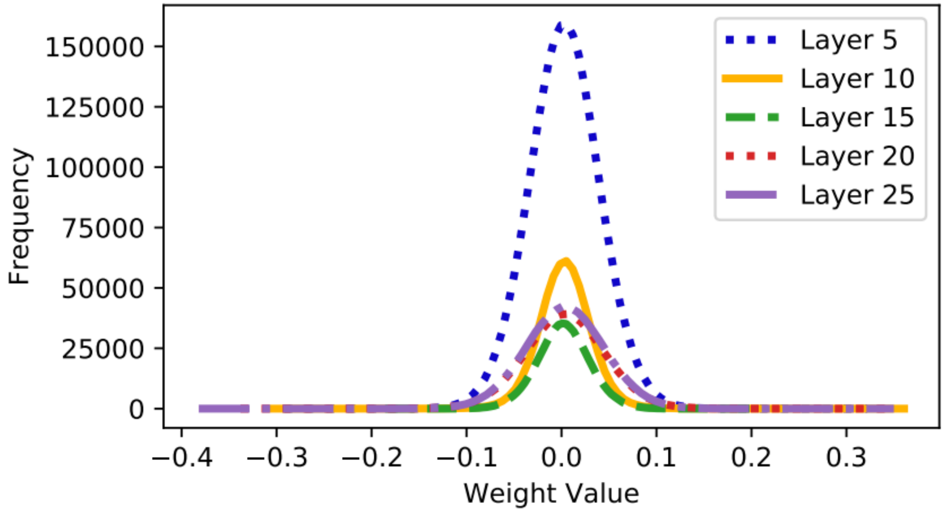


(a) per-tensor quantization

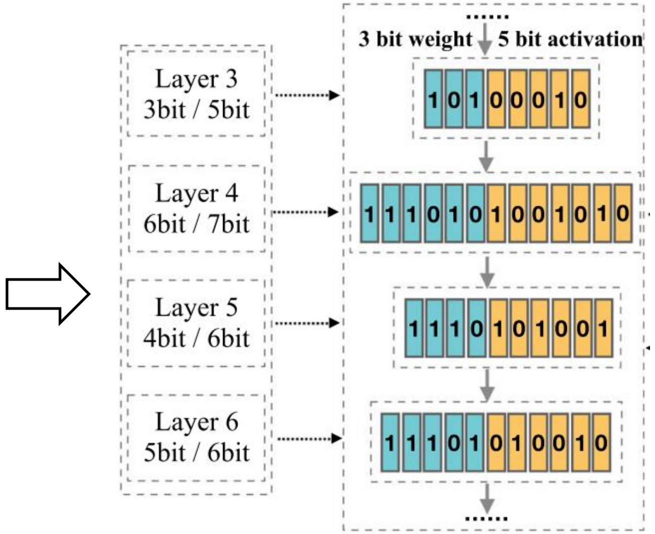
ZeroQuant: Efficient and Affordable Post-Training Quantization for Large-Scale Transformers, NeurIPS 2022

Mixed Precision Quantization

- Weights follow Gaussian distribution
- Outliers remain in original form, quantize the rest of the values
- Different bits for different layers



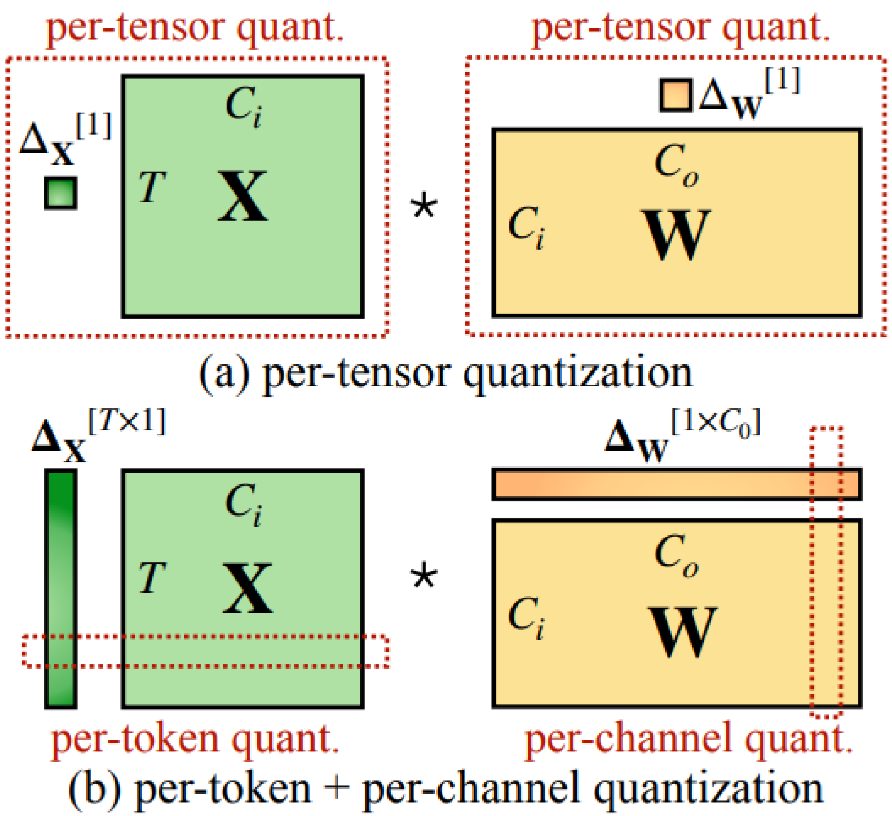
Per-layer weight distribution of BERT model



GOBO: Quantizing Attention-Based NLP Models for Low Latency and Energy Efficient Inference, MICRO 2020

Fine-grained Quantization

- Per-tensor quantization
  - Low accuracy
  - Fast to quantize/dequantize
- Per-token/channel quantization
  - High accuracy
  - Slower to quantize/dequantize
  - Custom kernels required



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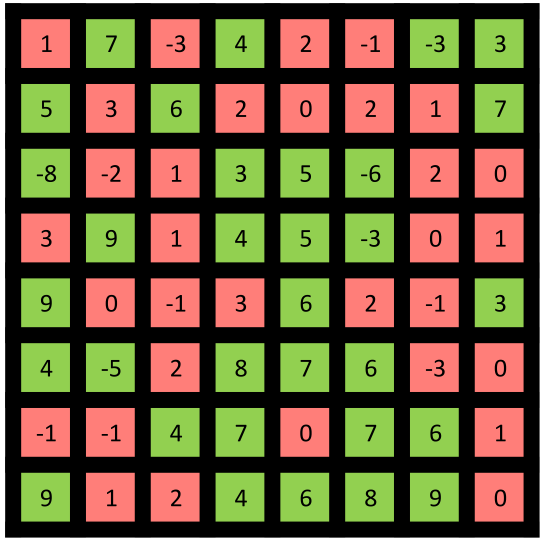
Sparsification

Unstructured (connection) Sparsity:

- High accuracy
- No performance improvement or performance regression



Unstructured Sparsity

SliceGPT: Compress Large Language Models by Deleting Rows and Columns, ICLR 2024

Sparsification

Unstructured (connection) Sparsity:

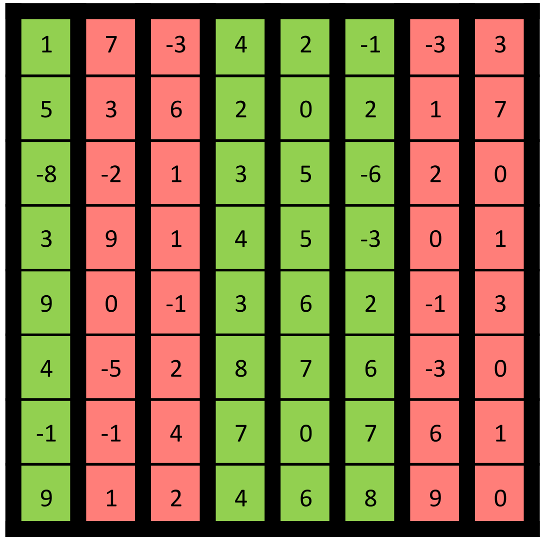
- High accuracy
- No performance improvement or performance regression

Structured Sparsity:

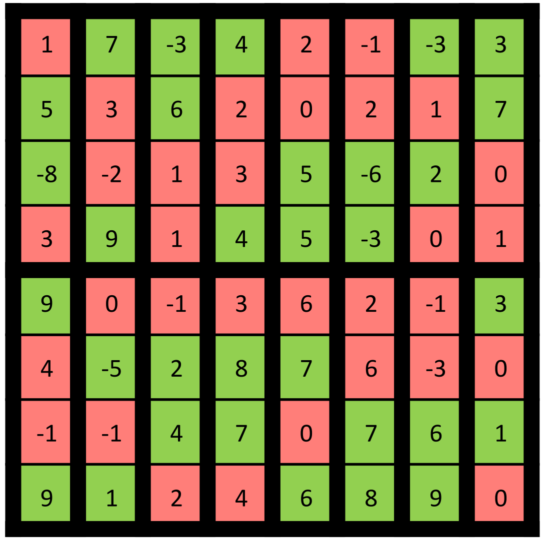
- Large accuracy degradation
- High performance scalability

N:M Semi-Structured Sparsity:

- High accuracy
- High performance improvement



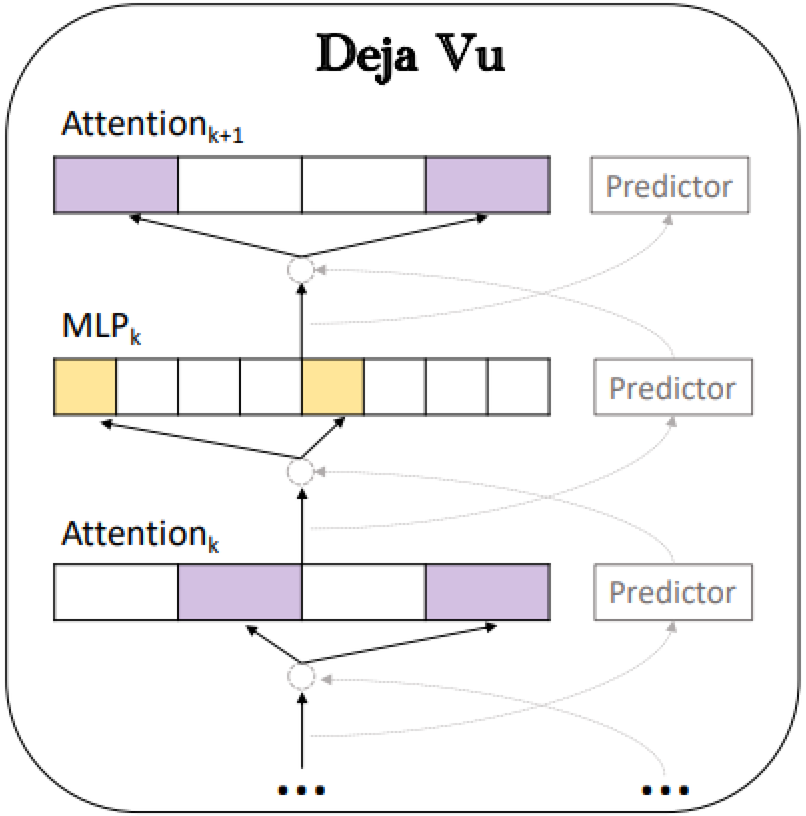
Structured Sparsity (Column-wise Sparsity)



Semi-Structured Sparsity (4:2 N:M)

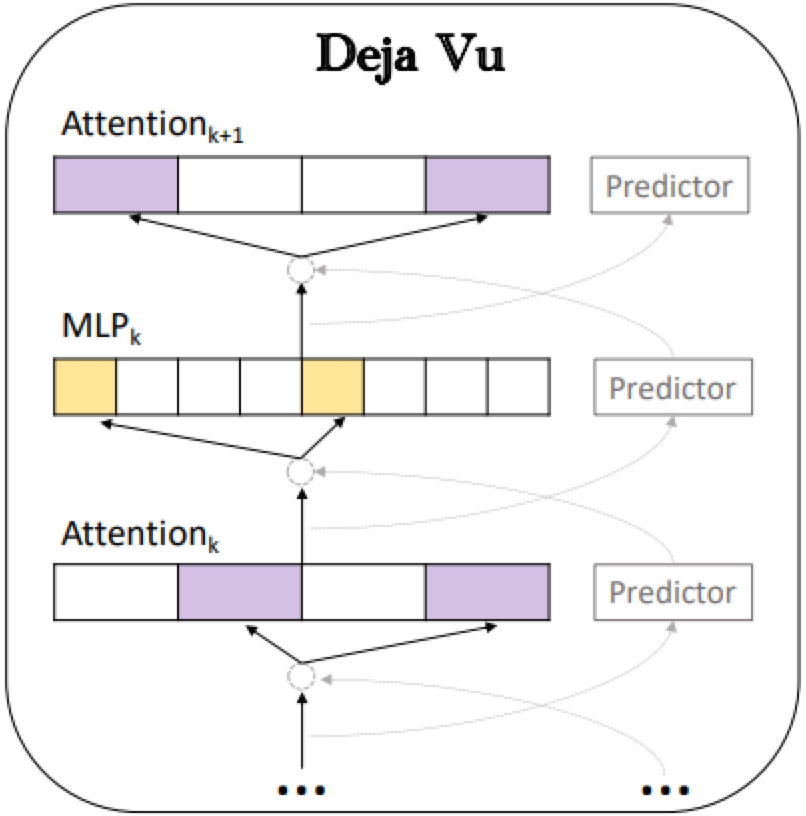
SliceGPT: Compress Large Language Models by Deleting Rows and Columns, ICLR 2024

Model Pruning



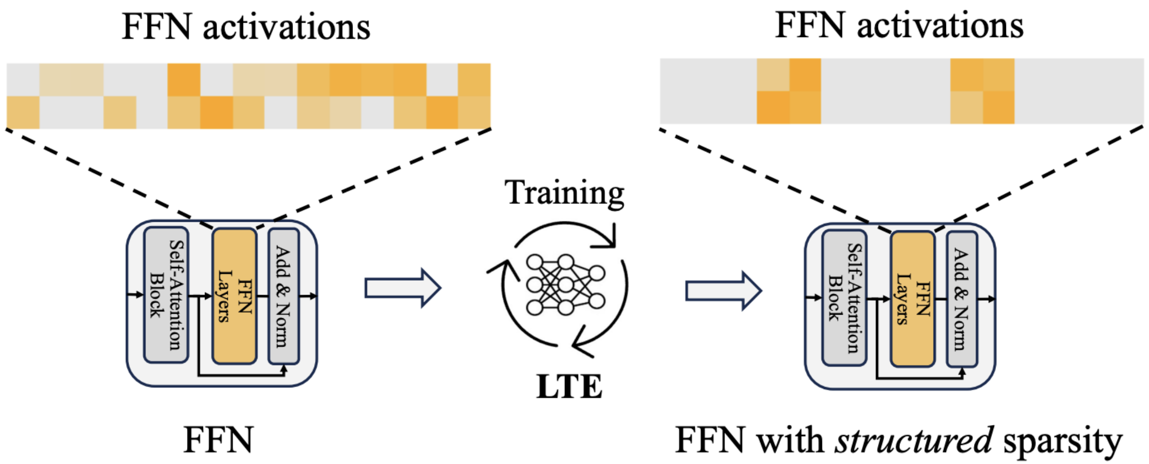
Deja Vu: Contextual Sparsity for Efficient LLMs at Inference Time, 2023

Model Pruning

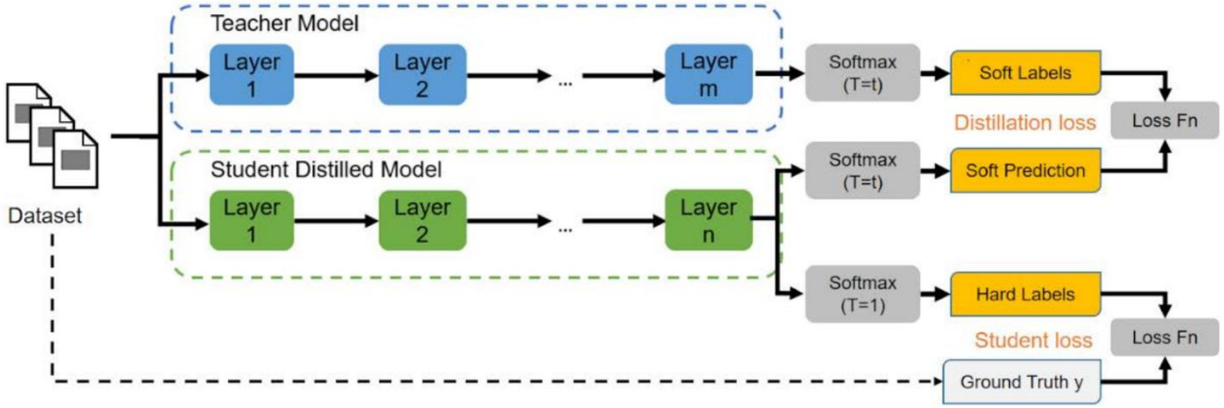


Deja Vu: Contextual Sparsity for Efficient LLMs at Inference Time, 2023

Learn To be Efficient: Build Structured Sparsity in Large Language Models, 2024

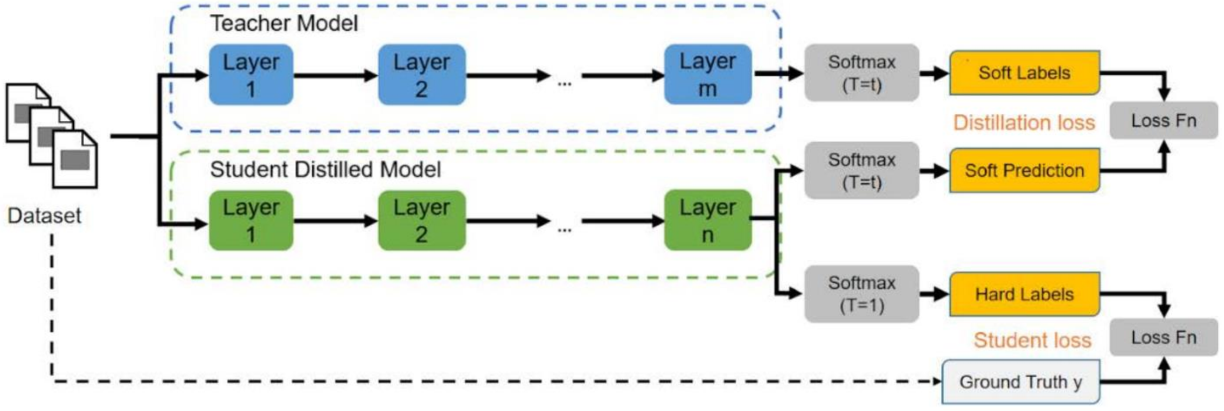


Knowledge Distillation



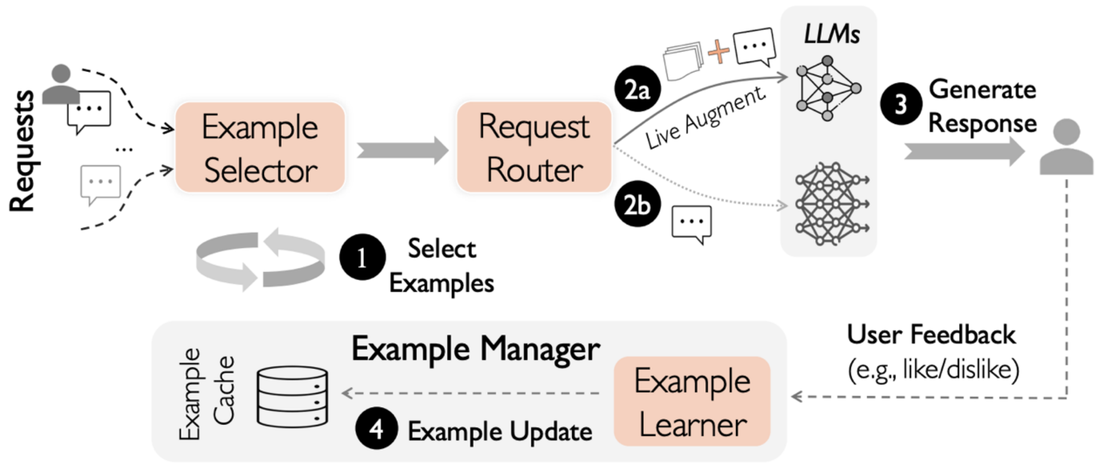
Distilling the Knowledge in a Neural Network
, 2015

Knowledge Distillation

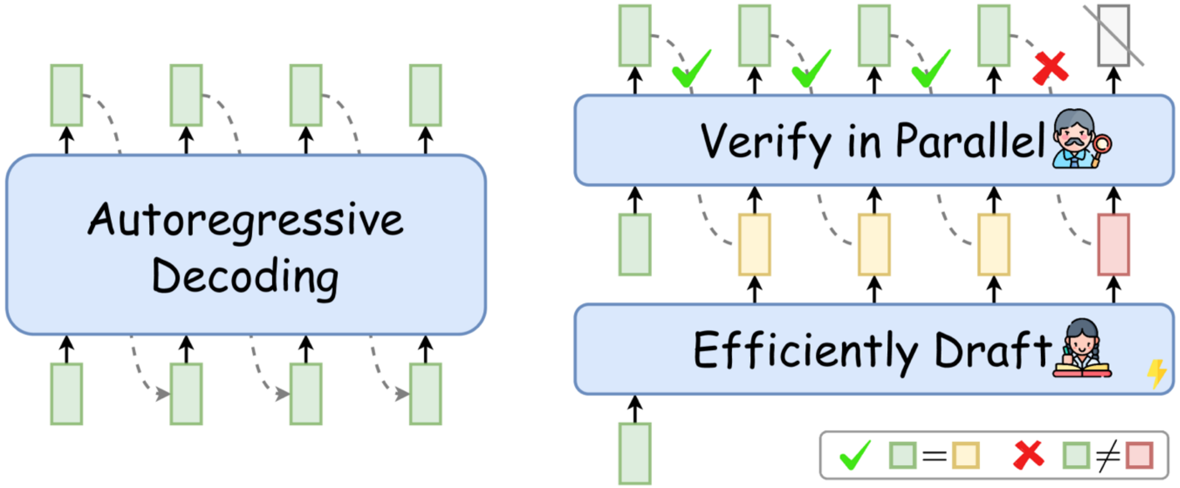


Distilling the Knowledge in a Neural Network , 2015

IC-Cache: Efficient Large Language Model Serving via In-context Caching, SOSP'25



Panel Discussion: Speculative Decoding



Fast Inference from Transformers via Speculative Decoding, 2023

Questions?